Require: design d, attacker a, legitimate traffic ℓ, seed z, attack minutes T, warm-up minutes W
Ensure: run record R

$(\mathrm{rng}_\ell, \mathrm{rng}_a, \mathrm{rng}_c) \leftarrow \textsc{Streams}(z)$
$P \leftarrow \textsc{BuildPipeline}(d)$; $\textsc{SeedReturningUsers}(P, 0.2)$
$E \leftarrow \emptyset$; $\mathrm{leaked}, \mathrm{offered} \leftarrow \mathbf{0}_T$
for $m \leftarrow -W$ **to** $T - 1$ **do**
    $n_\ell \sim \mathrm{Poisson}(\lambda_\ell)$; $n_a \sim \mathrm{Poisson}(\lambda_a)$ $[m \geq 0]$
    $A \leftarrow \textsc{Arrivals}(n_\ell, n_a, m)$ sorted by time
    for all $(t, \mathrm{kind}) \in A$ **do**
        $\textsc{ApplyEvents}(E, t)$
        $q \leftarrow \textsc{DrawRequest}(\mathrm{kind})$; $y \leftarrow P.\textsc{ProcessRequest}(q)$
        if $y = \mathrm{challenge}$ **and** $q.\mathrm{solves}$ **then**
            $y \leftarrow P.\textsc{ProcessRequest}(q \,\|\, \mathrm{solution})$
        end if
        if $y = \mathrm{sent}$ **then** $E \leftarrow E \cup \{\mathrm{receipt}(q), \mathrm{entry}(q)\}$
    end for
    $\textsc{ApplyEvents}(E, \mathrm{end}(m))$; $P.\textsc{ResolveTimeouts}()$
    $P.\textsc{AdaptiveJob}()$
    if $m \geq 0$ **then** $\mathrm{leaked}[m] \leftarrow \textsc{SentTo}(a, m)$; $\mathrm{offered}[m] \leftarrow n_a$
end for
$\textsc{ApplyEvents}(E, \infty)$
return $R \leftarrow (\mathrm{leaked}, \mathrm{offered}, \textsc{Funnels}(P), \textsc{Events}(P), d, a, \ell, z)$